PART 3: IN-CLASS EXERCISE

Logistic Regression vs. KNN

Two datasets, one question

Breast Cancer

Real classification data

569 observations
30 features
Malignant vs. benign

`make_moons`

Simulated nonlinear data

300 observations
2 features
Curved two-moon pattern

How do the same classifiers behave on different data structures?

Your Coding Task

Complete five TODOs in the starter file

1. Build Logistic Regression.
2. Set KNN to $k = 5$.
3. Make a stratified 70/30 train-test split.
4. Fit fresh copies of both models.
5. Compare both datasets.

You have about four minutes.

Use the same workflow twice

```
StandardScaler()
  ↓
LogisticRegression(max_iter=1000)
```

```
StandardScaler()
  ↓
KNeighborsClassifier(
  n_neighbors=5)
```

Breast Cancer: accuracy + malignant recall

`make_moons`: accuracy + boundary shape

Breast Cancer: Accuracy Is Not Enough

Model	Test accuracy	Malignant recall
Logistic Regression	**0.988**	**0.984**
KNN ($k = 5$)	0.959	0.891

Discussion

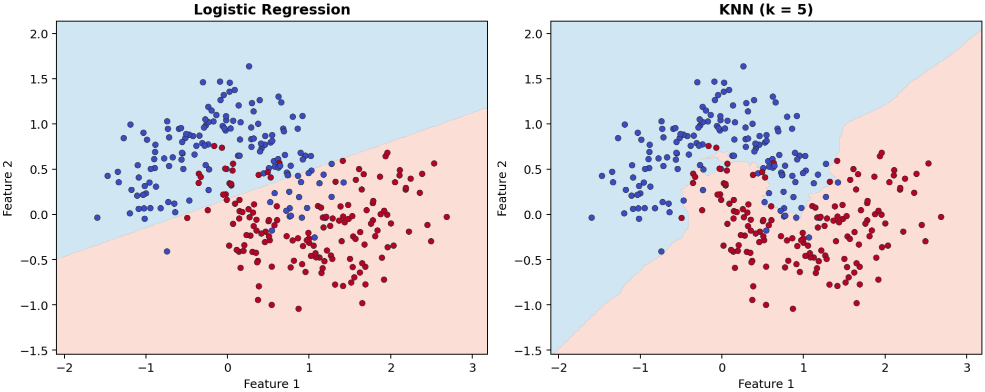
Is the model with the highest accuracy always the safest model?

Error cost matters

False negative: malignant $\rightarrow$ benign
Malignant label is `0`, so use `pos_label=0`.

In medical classification, recall can matter as much as accuracy.

`make_moons`: Boundary Shape Matters



Logistic Regression

Linear boundary: accuracy $= 0.856$

KNN $(k = 5)$

Flexible local boundary: accuracy $= 0.911$

Choose models using data structure, evaluation goals, and error cost.